

# Cervical Cytology-Histology Correlation Quality Assurance Report

Automated pipeline per ASC Birdsong Guideline 2017 | CAP CYP.06600

Total Cases Analyzed: 130

## Executive Summary

A total of 130 cases were analyzed.

**CAP ALERT:** Major discordance rate of 31.5% exceeds the CAP benchmark of less than 10%. Corrective action documentation required per CYP.06600.

**PV+ ALERT:** HSIL positive predictive value of 27.3% falls below the CAP target of 60%.

## Section 1 ? Concordance Summary

Classification of all cytology-histology pairs per ASC Birdsong concordance definitions.

| Concordance_Class | Count | Percent |
|-------------------|-------|---------|
| minor_discordant  | 51    | 39.23   |
| major_discordant  | 41    | 31.54   |
| concordant        | 33    | 25.38   |
| unmapped          | 5     | 3.85    |

### Agreement Within One Grade (Birdsong Section 5)

Percentage of pairs with exact agreement or minor disagreement only. Per Birdsong, this metric is of educational interest.

| Metric                                | Count | Percentage |
|---------------------------------------|-------|------------|
| Agreement within one grade            | 84    | 64.6%      |
| Exact agreement only                  | 33    | 25.4%      |
| Major discordance (outside one grade) | 41    | 31.5%      |

## Section 2 ? Discrepancy Bucket Summary

CAP-style discrepancy classification per Birdsong Figure 1 and Figure 3.

| Bucket  | Count | Percent |
|---------|-------|---------|
| MinVar  | 4     | 3.08    |
| MajUnd  | 27    | 20.77   |
| MinUnd  | 31    | 23.85   |
| Agree   | 33    | 25.38   |
| MinOver | 16    | 12.31   |
| MajOver | 14    | 10.77   |

## Section 3 ? HSIL-Focused Metrics (Birdsong Section 7)

The five statistical calculations required by Birdsong Section 7 and the ASC Clinical Practice Committee.

| Metric                                          | Value |
|-------------------------------------------------|-------|
| HSIL Pap tests with histologic HSIL (CIN2/CIN3) | 27.27 |

| Metric                                                            | Value |
|-------------------------------------------------------------------|-------|
| HSIL Pap tests with minor discrepancies                           | 45.45 |
| HSIL Pap tests with major false-positive discrepancies            | 27.27 |
| Number of major false-negative / cytology undercall discrepancies | 27.0  |
| PV+ for HSIL Pap tests (%)                                        | 27.27 |

#### ***PV+ Interpretation (Birdsong Section 7a)***

**LOW ? Below CAP target of 60%. Review HSIL overcall rate.**

#### ***Extended PV+ ? High-Grade Cytology Group (HSIL + ASC-H + AGC-NEO)***

Per Birdsong Figure 1, HSIL, ASC-H, and AGC-NEO are grouped together for concordance assessment. The extended PV+ reflects combined performance.

| Metric                                              | Value |
|-----------------------------------------------------|-------|
| Total high-grade Pap tests (HSIL + ASC-H + AGC-NEO) | 56    |
| High-grade Pap tests with HSIL+ histology           | 35    |
| Extended PV+                                        | 62.5% |

## **Section 4 ? Intergrade Follow-Up Rates (Birdsong Section 8)**

Per Birdsong Section 8: review of HSIL histology rates following ASC-US, ASC-H, and LSIL cytologic interpretations to identify systematic undercall patterns.

| Cytology | Total Cases | HSIL+ Follow-up | HSIL+ Rate (%) |
|----------|-------------|-----------------|----------------|
| ASC-US   | 13          | 9               | 69.2           |
| ASC-H    | 9           | 7               | 77.8           |
| LSIL     | 15          | 11              | 73.3           |
| AGC      | 4           | 2               | 50.0           |
| AGC-NEO  | 9           | 8               | 88.9           |

Elevated HSIL+ follow-up rates in ASC-US or LSIL categories may indicate systematic cytologic undercalling requiring targeted slide review per Birdsong Section 8.

## **Section 5 ? Key Quality Signals**

| Quality Signal                    | Value | CAP Threshold | Status  |
|-----------------------------------|-------|---------------|---------|
| Major discordance rate            | 31.5% | < 10%         | EXCEEDS |
| Minor discordance rate            | 39.2% | ?             | ?       |
| Undercall proportion              | 44.6% | < 15%         | EXCEEDS |
| Overcall proportion               | 23.1% | ?             | ?       |
| HSIL PV+                          | 27.3% | > 60%         | LOW     |
| Extended PV+ (HSIL+ASC-H+AGC-NEO) | 62.5% | > 60%         | ?       |
| Agreement within one grade        | 64.6% | ?             | ?       |
| Major false negative count        | 27    | Minimize      | REVIEW  |

Section 6 ? Visualizations

Figure 1 ? Concordance Distribution

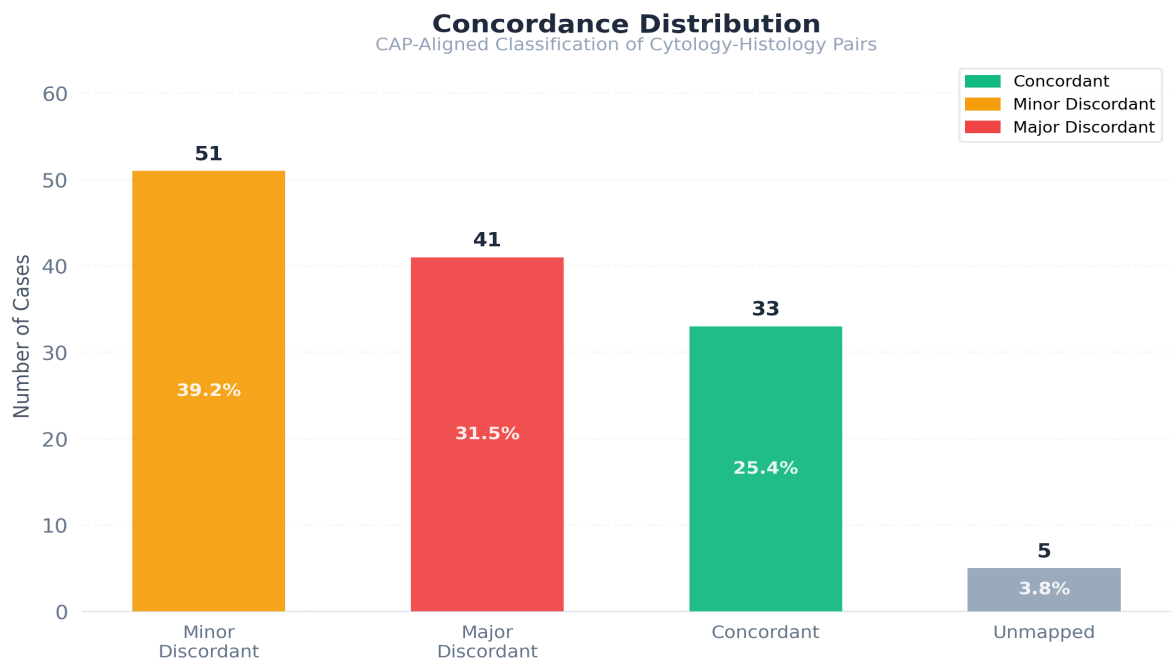


Figure 2 ? Cytology-Histology Correlation Buckets

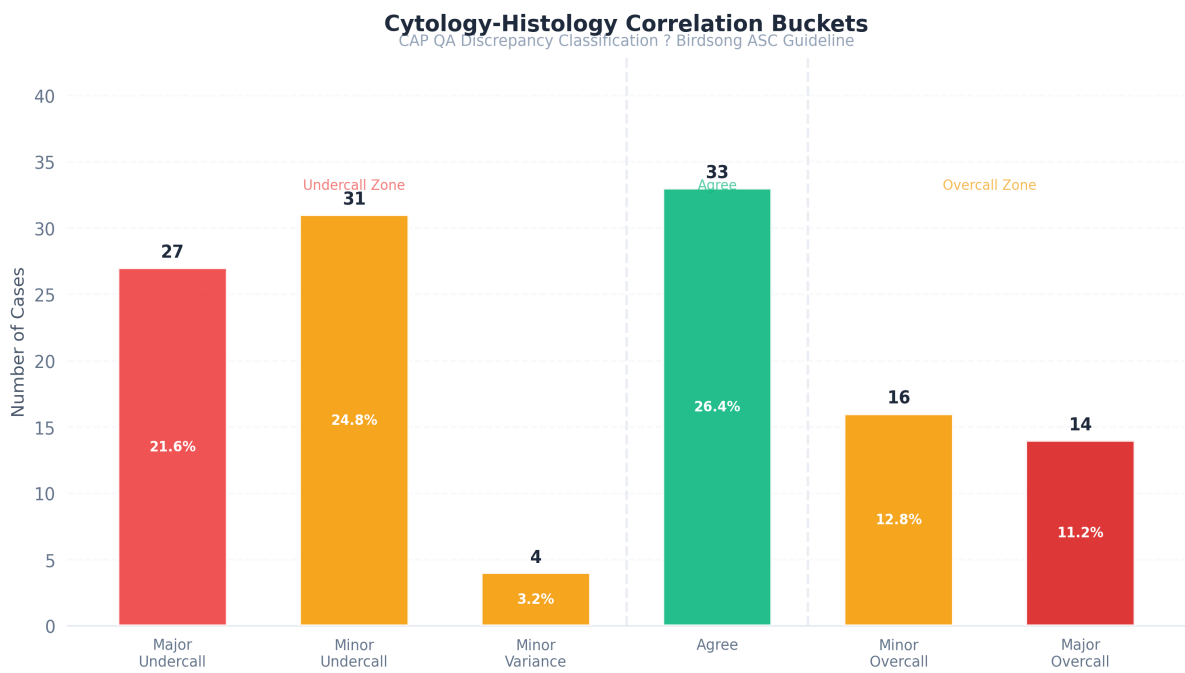


Figure 3 ? HSIL Correlation Metrics

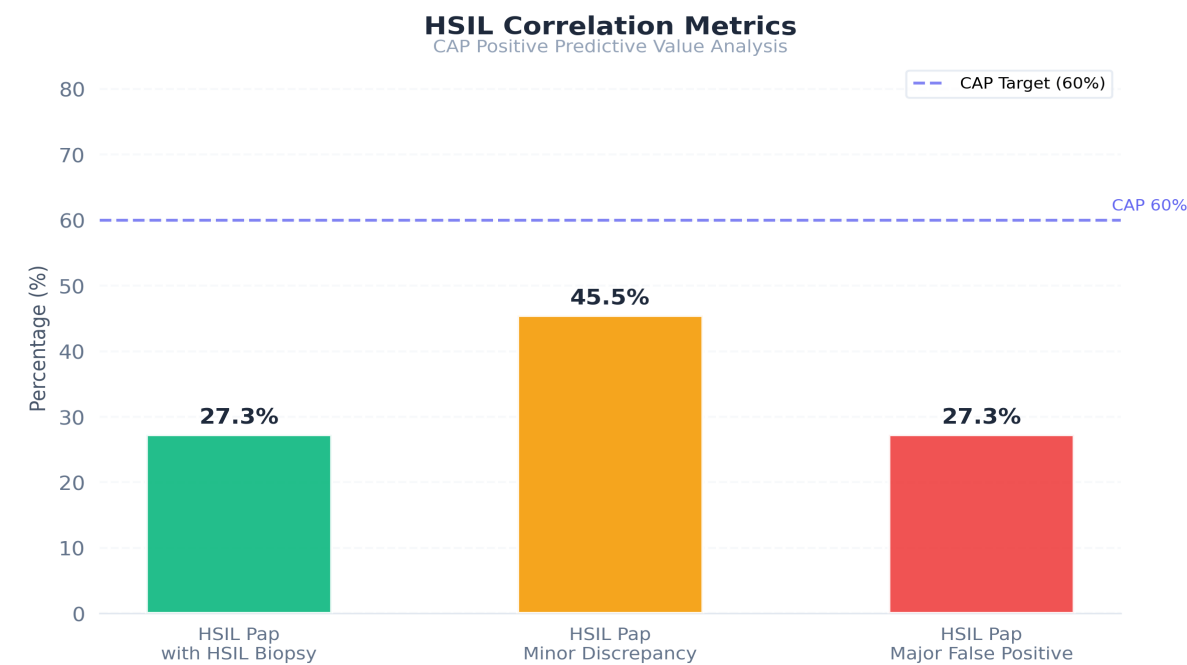


Figure 4 ? Cytology vs Histology Confusion Matrix

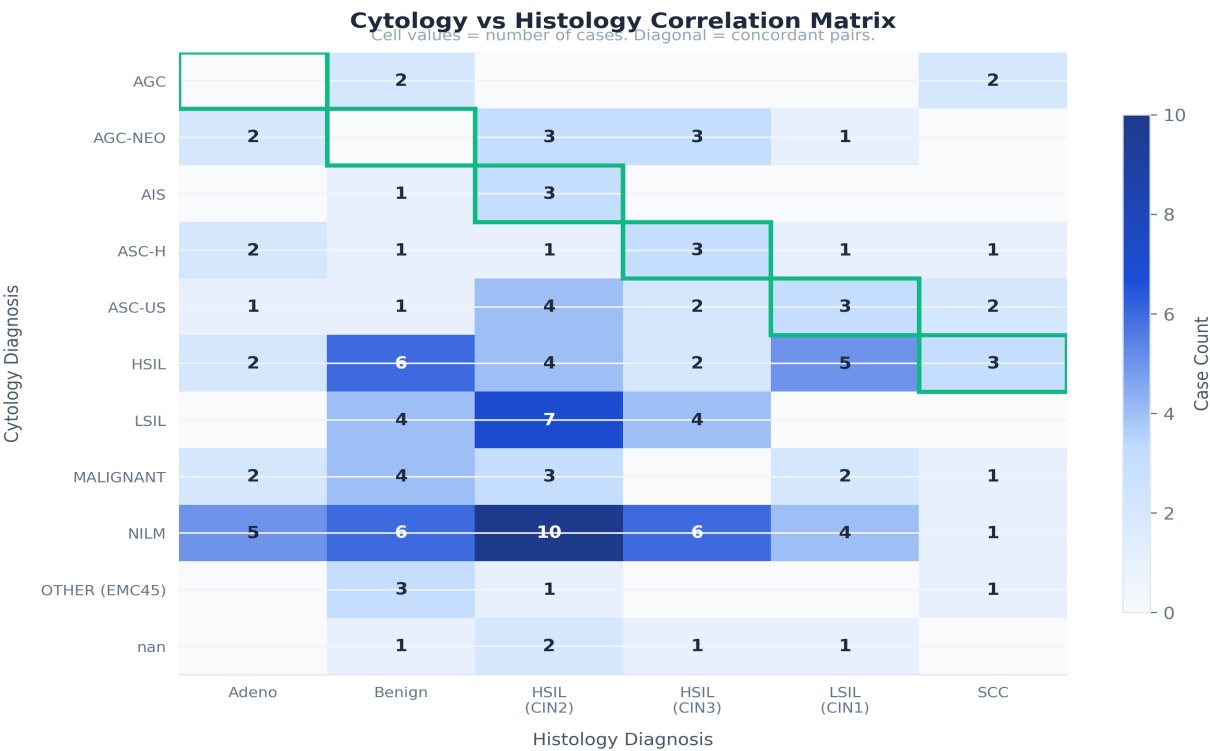
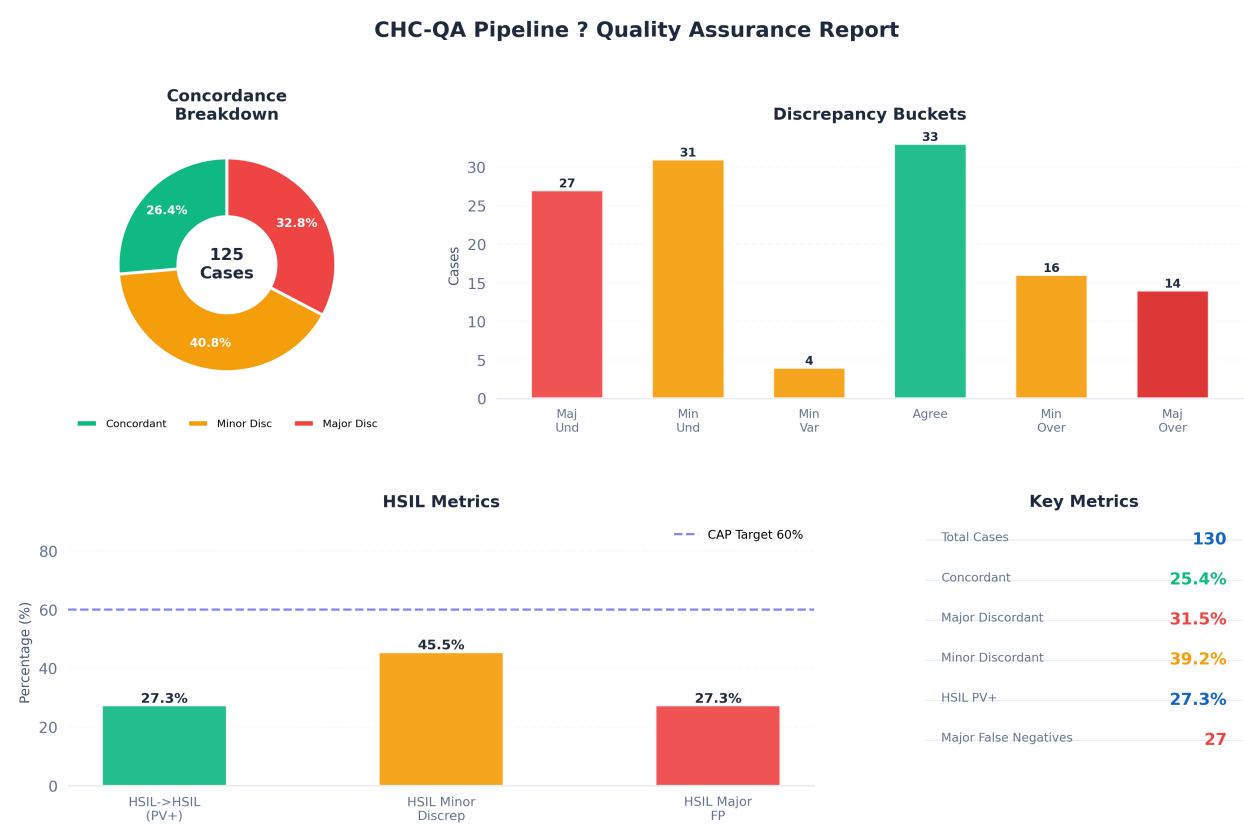


Figure 5 ? Summary Dashboard



## Section 8 ? Limitations

This report is generated by an automated deterministic cytology-histology correlation pipeline. Classification rules are derived from the ASC Clinical Practice Committee Birdsong Guideline (2017) and applied deterministically to all pairs. The following limitations apply:

1. Verification bias: CHC datasets are subject to verification bias due to selective histologic follow-up. Results should not be interpreted as measures of screening sensitivity or specificity.
2. Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.
3. LLM normalization: Free-text diagnosis normalization was performed by a locally-hosted large language model. Normalized terms were validated against a controlled vocabulary. Residual normalization errors may affect a small number of cases.
4. Pairing logic: Cases were paired by patient identifier and date window. Complex scenarios including multiple biopsies, multiple Pap tests, and delayed follow-up may result in suboptimal pairing in a minority of cases.
5. Root cause classification: This pipeline classifies discordance type and direction but does not perform root cause analysis (sampling error, screening error, interpretive error). Manual review of major discordant cases is required per Birdsong Section 8.

Reference: Birdsong GG, Walker JW. Gynecologic Cytology-Histology Correlation Guideline. ASC Bulletin. 2017;LIV(2):VIII-XIII.